

Requirement:

1. Select proper datasets from different categories of those publically available (e.g., from UCI, Kaggle, etc.) to experiment with association analysis, classification, and clustering.
2. Using scikit-learn, experiment with different ML algorithms and report your findings detailing the performance of each as you run your experiment.
3. Explain your validation process.
4. Tune the associated hyperparameters,
5. Select the best performing versions and make comparisons among the different algorithms using performance assessment methods as well as the time required.
6. Compare your work with that in the literature using the same datasets.
7. **Make sure to have proper UI to conveniently lead the users to run and use your system.**
8. For classification, experiment with at least 10 different algorithms (techniques).
9. For others, experiment with at least 2-3 different algorithms (techniques).
10. Submit a report detailing your work. Follow the report instructions blow.
11. Upload the report word file and the presentation ppt files into the LMS. In the report, describe all the steps to design and implement your models including the platform (**SKlearn**) you used to develop the models, the algorithms including employed hyperparameters, the data preprocessing you applied to the dataset, the evaluation and comparison results in tables and graphs.
12. Highlight major observations and make conclusions.
13. Give a Demo presentation.
14. Upload a ZIP archive for the source code, a readme file, and screenshots of all the code & evaluation results from the platform you used.
15. Upload a video explaining your work.

Evaluation criteria:

1. Completeness
2. UI
3. Demo and presentation
4. Write up clarity and correctness

Report instructions:

- a. Make sure the writing is in your own words with references throughout the text when you are addressing issues or results from the literature.
- b. The abstract should include: Focus, objectives, methodology, result, and conclusion in about 300 words.

- c. The Introduction should indicate the objectives of the study and provide enough background information to clarify why the study was undertaken and what hypotheses were tested.
- d. Followed by sections about your tool including related work, usage, applications, comparison, experiments, etc.
- e. A conclusion of your work and finding.
- f. List of references. Write your references and in-text citations using IEEE citation style. You may want to look at <https://libguides.murdoch.edu.au/IEEE/home>
- g. Use times roman 11pts for text and 12 pts bold for section titles. Use 1.5 spacing.
- h. Prepare 10 ppt slides from your report to present in class. Be informative and concise.
- i. Make up 5 questions that the listeners can answer.
- j. Submit soft copies in word and ppt of the report, questions, and presentation.